AIAA 3201 · PROJECT 4 · PART 1

Initialization, not architecture

COLMAP vs. VGGT-1B vs. VGGT-Ω as 3D Gaussian Splatting initialization under a controlled dense view budget

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What we compared, and how

FOUR AXES, AS SPECIFIED IN THE BRIEF

Initialization runtime
seconds to a training-ready scene

Pose accuracy
ATE after Sim(3) alignment; relative rotation

3DGS convergence speed
test PSNR every 5 k iterations to 30 k

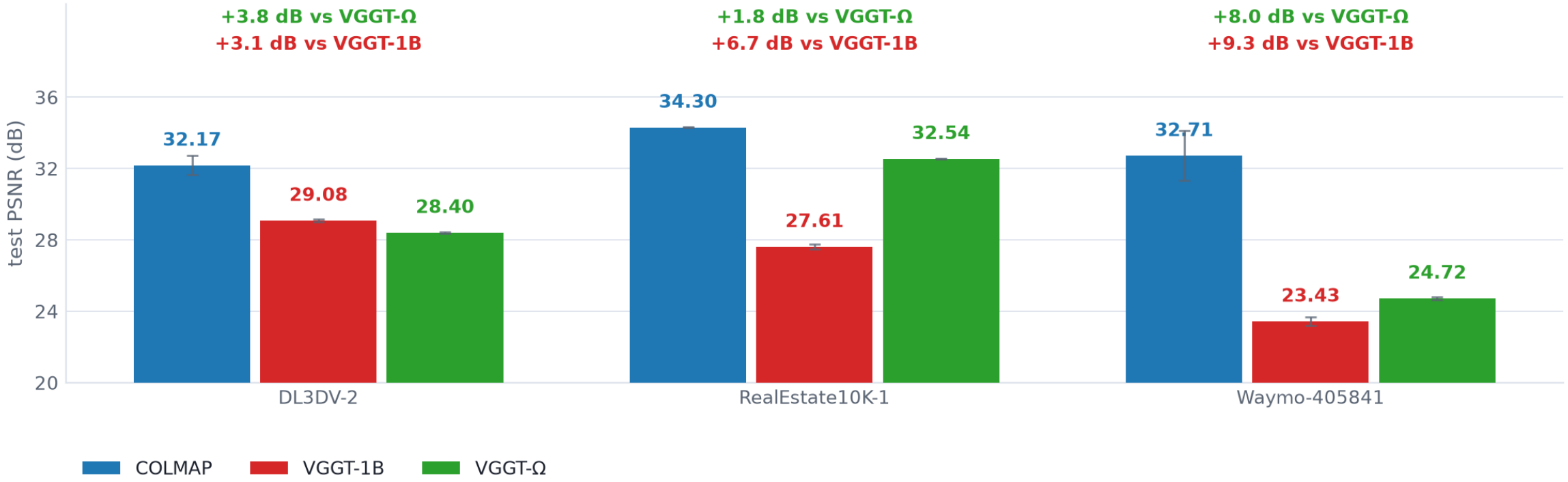
Final rendering quality
PSNR / SSIM / LPIPS on frozen test views

THREE MANDATORY SCENES

Scene	Type	Images	Test
DL3DV-2	outdoor	306	39
RealEstate10K-1	indoor	279	35
Waymo-405841	driving	199	25

- N = 160 training views — the dense regime; 8–17× denser than Part 2's sparse settings
- Splits frozen, llffhold = 8 — all three backends see the same images
- 30 000 iterations, densify to 15 000, reference hyper-parameters
- 2–3 seeds per configuration, with pairwise t-tests

COLMAP produces the best renders on all three scenes



y-axis truncated at 20 dB · error bars = run-to-run std (2–3 seeds)

Largest gap on Waymo-405841 (+9.3 dB). SSIM and LPIPS rank the backends identically, so the ordering is not an artefact of one metric.

The ordering is visible without metrics



Column headers give the scene-level mean test PSNR (mean over all test views, 2–3 seeds where available); corner labels give that single view's PSNR.
Note: the official VGGT exporter caps the point cloud at 100 000 points chosen by a random unseeded subsample (vggt/utils/helper.py::randomly_limit_trues), so VGGT-1B figures carry an export-level spread far larger than its training-seed spread — see the report's limitations.

WHAT TO LOOK AT

VGGT-1B, 21.4 dB

loses the fine structure of the red foliage and the paving joints.

VGGT-Ω, 28.3 dB

keeps the shape but drifts in colour — visibly greyer and colder.

COLMAP, 32.7 dB

closest to ground truth on all three rows.

ONE THING TO BE CAREFUL ABOUT

These renders use a different VGGT-1B export than the 29.1 dB in the table. The exporter subsamples the cloud with an unseeded **np.random.choice**, so that column moves by several dB between exports of the same model.

The mechanism is not pose accuracy

POINT CLOUD, DL3DV-2 (model_analyzer)

Model	Points	Mean track length
COLMAP	3 778	**8.19**
VGGT-1B	100 000	1.000
VGGT-Ω	150 000	1.000

TRACK LENGTH = 1.000 MEANS EVERY 3D POINT IS SEEN IN EXACTLY ONE IMAGE

— Both exporters emit per-frame depth unprojections, not multi-view triangulations.

— 3DGS therefore starts from mutually unconstrained samples and must reconcile their disagreements at render time.

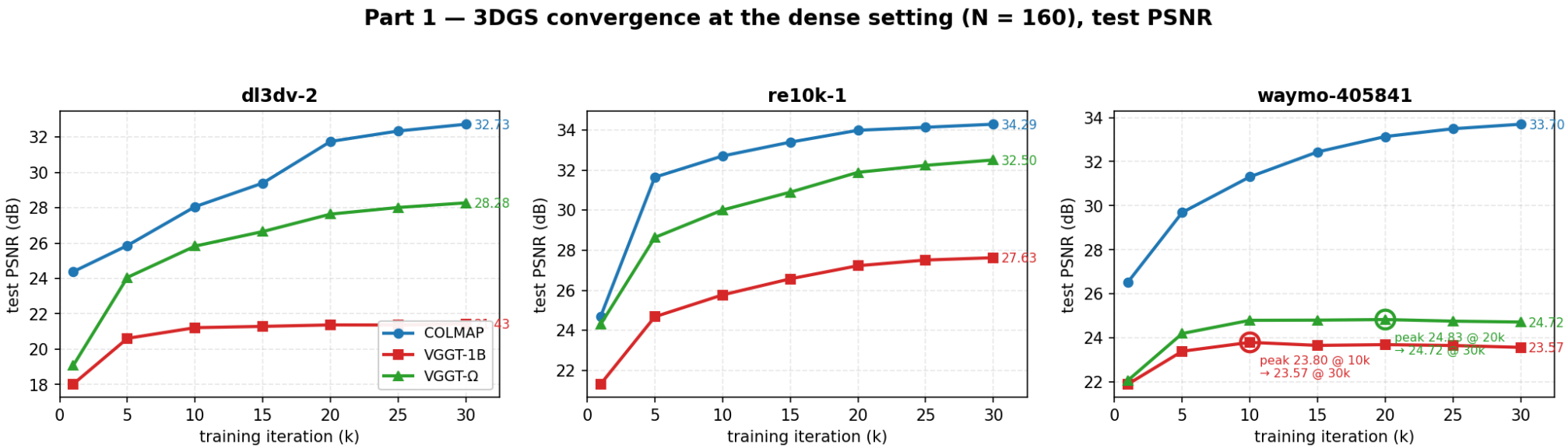
— The reported 0.000 px reprojection error is vacuous: a point seen once has no cross-view residual.

The contradiction, at 50 views (Re10k-1):
VGGT-Ω has **5× COLMAP's pose accuracy** (ATE 0.220 % vs 1.065 %) and still renders **1.35 dB worse**.

SO WHAT SHOULD YOU CONCLUDE?

- Pose accuracy alone cannot explain rendering quality.
- The feed-forward models are not worse reconstructors — they deliver competitive poses and dense but unconstrained geometry.
- Follow-up we did not run: keep VGGT's poses, re-triangulate points from multi-view correspondences.

Convergence: 30 k is the right budget at 160 views



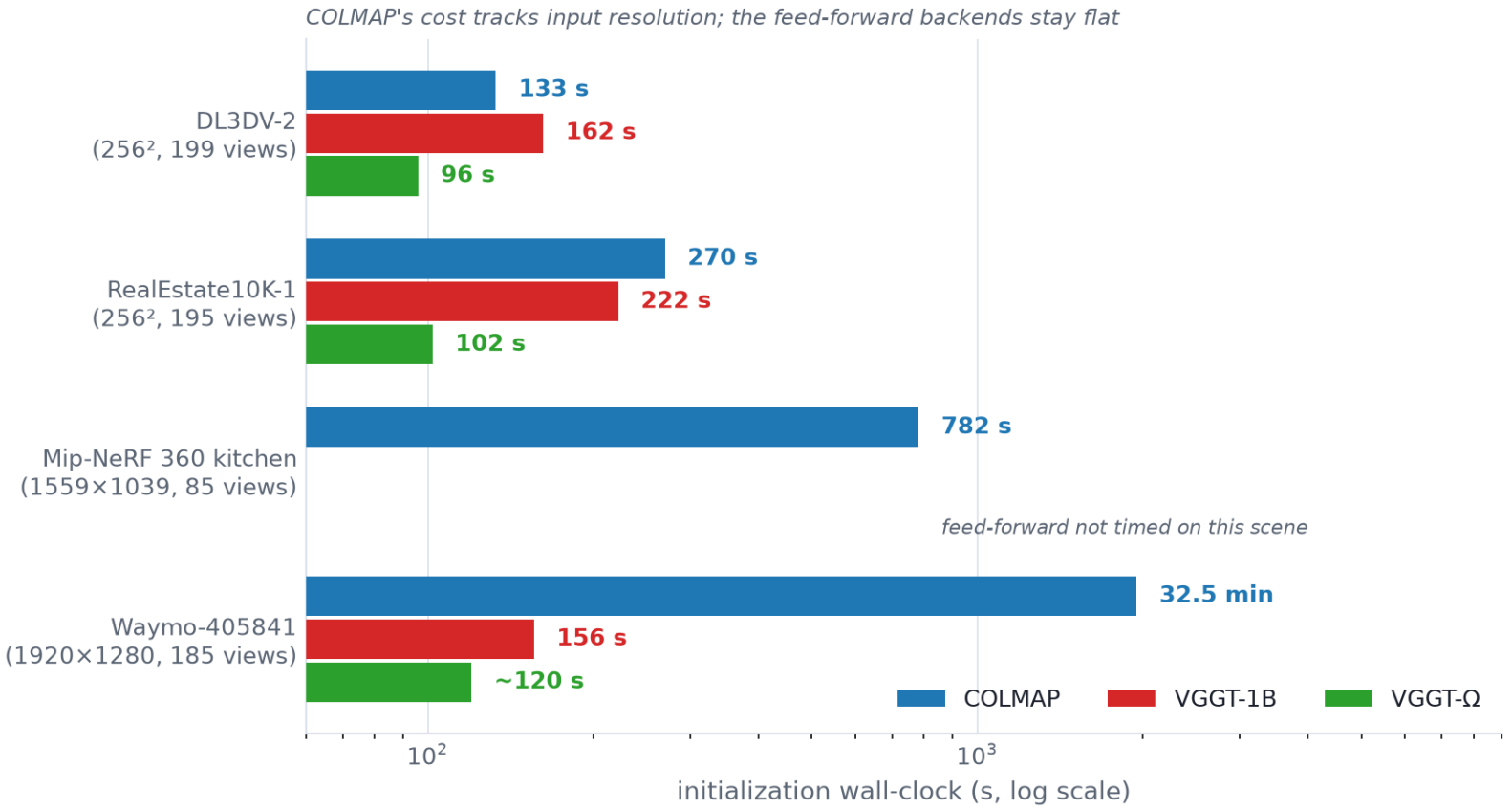
Hollow circles mark configurations whose peak occurs before 30 k and then decays; the two numbers give peak and final test PSNR.

7 of 9 configurations are still improving at 30 k — and all three COLMAP curves are.

Waymo is the exception: both feed-forward models peak early (10 k, 20 k) and then decay; COLMAP does not.

The "peak at 1 k" from the N=50 run does not occur at 160 — it was under-training, not over-fitting.

Speed and run-to-run stability



READING THE CHART

— Feed-forward: 70–220 s at any resolution — every input is normalized to a fixed 512² window.

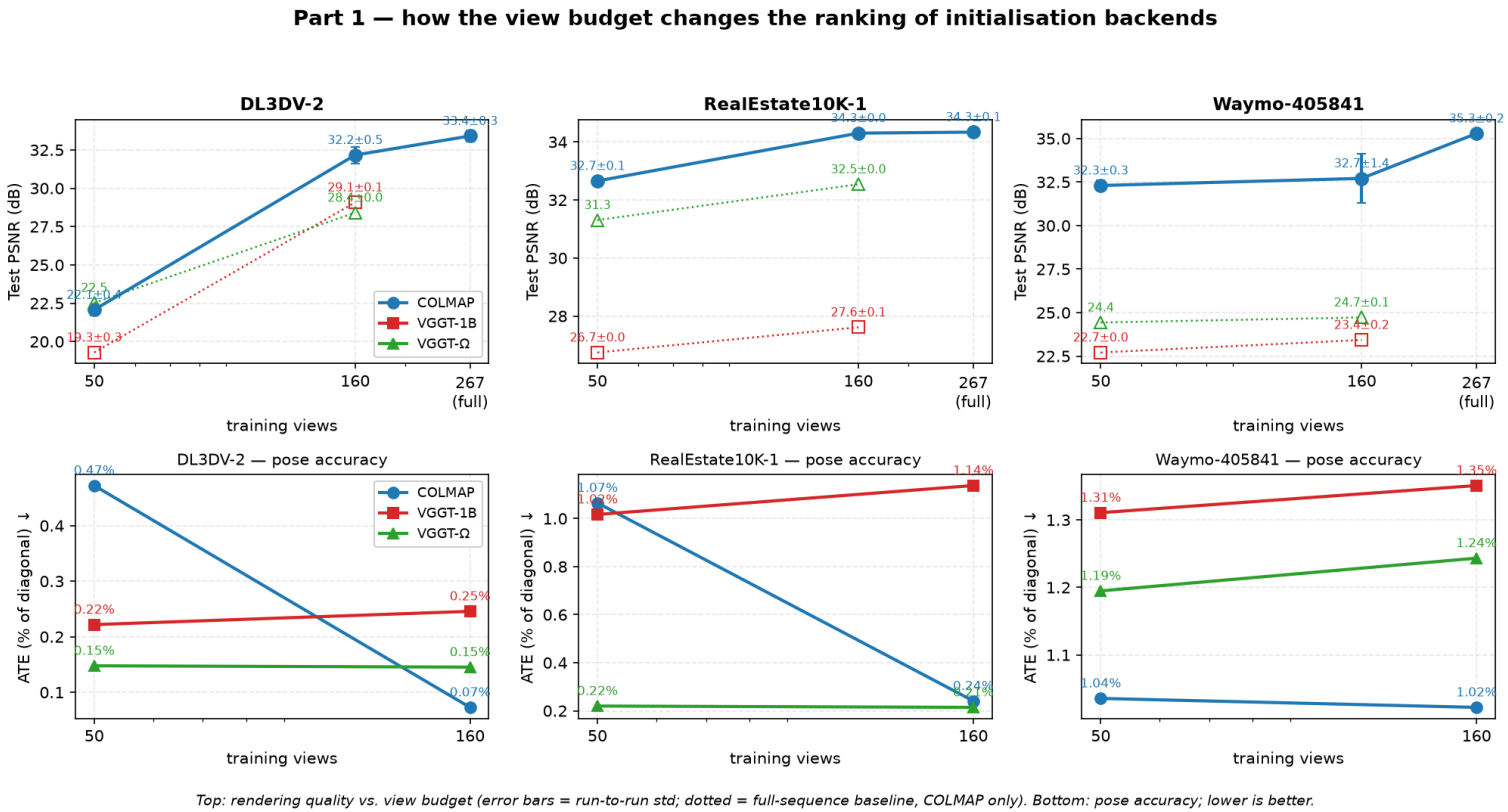
— COLMAP: 133 s → 1 950 s as resolution grows from 256² to 1920×1280.

— kitchen: 782 s on only 85 views — 5.9× DL3DV's cost at a quarter of the frames.

AND IT REPRODUCES

- **8 of 9 configurations** reproduced to within 0.12 dB on a second GPU, rebuilt from scratch.
- **The one outlier:** VGGT-1B's exporter subsamples the cloud with an unseeded np.random.choice.

The gap widens as observations accumulate



Top: rendering quality vs. view budget (error bars = run-to-run std; dotted = full-sequence baseline, COLMAP only). Bottom: pose accuracy; lower is better.

COLMAP's pose error collapses 6.5× (DL3DV-2) and 4.4× (Re10k-1) from 50 → 160 views; feed-forward error does not move. VGGT-Ω's deficit over COLMAP grows on all three scenes, which is what moves it from first place at 50 views to last at 160.

Bonus: it also holds on Mip-NeRF 360

THE 3DGS PAPER'S OWN BENCHMARK — OPTIONAL IN THE BRIEF

Scene	Images	COLMAP	VGGT-1B	Gap
counter	240	**29.47**	23.66	+5.81
bonsai	292	**31.27**	17.03	**+14.24**
flowers	173	**21.49**	15.51	+5.99

COLMAP is re-run by us on the same frames, not the bundled all-frame model — otherwise the comparison would not be like-for-like.

WHAT IT SHOWS

+5.8 to +14.2 dB. The COLMAP advantage is the same order as on the three mandatory scenes, on a benchmark we did not tune on.

bonsai is the strongest form of the scaling asymmetry. Going from 50 to 160 views, COLMAP improves (28.33 → 31.27 dB) while VGGT-1B gets **worse** (19.05 → 17.03 dB).

Caveats. Six configurations, n = 1, no significance test. VGGT-Ω was not run here. kitchen is absent from the weights mirror we used, so flowers was substituted.

3 of 3 scenes, same direction, no reversal — the optional-deliverable check passes, and on a benchmark we never tuned on.

Seven silent failure modes

Every number above required eliminating these first. None produced an error message. Two individually exceeded the effects we report as findings.

#	Failure	Silent symptom	Impact
1	Diverged COLMAP camera	camera centre at 7.2×10^{9}; scene box 7.2×10^{9}	**+13.0 dB**
2	Double-normalized focal in the Ω export	exported fx = 0.4 instead of 204	**+13.4 dB**
3	Mixed alignment procedures	rotation error inflated 20× on a near-planar path	would mis-rank
4	Wrong Waymo GT convention	rotation error 23.5° instead of 1.38°	poses look broken
5	Dataset poses hide SfM cost	initialization time = 0	speed axis impossible
6	Missing sparse/0 directory	"Could not recognize scene type!"	pipeline won't start
7	LPIPS weights fetched at eval time	528 MB at 150 KB/s	~1 h of stalls

Failures 1 and 2 are the same species: scale errors that leave the optimizer running happily, detectable only by checking an intermediate quantity against a known reference. Watching the loss curve would not have caught either.

Takeaways

> **COLMAP-initialized 3DGS renders best on all three scenes, and the margin widens as observations accumulate.**

- **Pose accuracy is not the explanation.** COLMAP's error falls 4–6× from 50 → 160 views while feed-forward error stays flat — and at 50 views COLMAP has the worst poses yet the best renders.
- **VGGT-Ω is the best feed-forward option.** By far the most reproducible (std 0.03–0.07 dB), and it beats COLMAP at 50 views on DL3DV-2 — but it gains least from more views and falls to last place.
- **The decisive variable is point-cloud topology.** Track length 8.19 for triangulation vs 1.000 for per-frame depth unprojection.
- **What data-driven scaling buys:** a 70–100 s, resolution-insensitive, highly reproducible initializer — at the cost of 3–9 dB of fidelity.

Limitations: VGGT-1B's export is stochastic (±several dB); Mip-NeRF 360 is single-seed; no hybrid VGGT-poses + re-triangulated-points run; COLMAP timings are CPU-bound.

Nine flythroughs, one per scene and budget

dense · mandatory

DL3DV-2 · N = 160



RealEstate10K-1 · N = 160



Waymo-405841 · N = 160



dense · Mip-NeRF 360

M360 counter · N = 160



M360 bonsai · N = 160

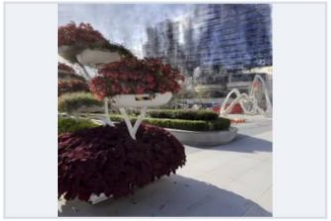


M360 flowers · N = 160



sparse · N = 50

DL3DV-2 · N = 50



M360 counter · N = 50



M360 kitchen · N = 50



Each clip: 8 s flythrough at 30 fps, rendered from the same evaluation pass that produced the tables. Frame 120 shown.

Six dense clips and three at N = 50, every one 8 s at 30 fps. Rendered from the trained models by the same evaluation pass that produced the PSNR tables, so the videos and the numbers are the same artefacts. Package: docs/videos/, nine files, 69.6 MB.

Animated — Waymo-405841 at 160 views



WHAT TO WATCH

The far kerb

shimmers and re-forms as the camera moves — the signature of geometry that is dense but mutually unconstrained.

Parked cars on the left

keep their silhouette but lose their edges; this is the scene where the gap between backends is largest (+9.3 dB).

Not a still

the trajectory is continuous, so temporal consistency is on display, not just per-frame fidelity.

▶ **Loops automatically in slide show.** The other eight clips are separate mp4s in docs/videos/; embedding all nine would add ~70 MB.